



Machine learning prediction of groundwater arsenic contamination using water quality parameters in the coastal region of Bangladesh

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Abstract Groundwater arsenic contamination poses a significant health risk in coastal region of Bangladesh. However, existing studies have rarely applied advanced machine learning (ML) algorithms to predict arsenic concentrations using comprehensive water quality parameters (WQPs). Thus, this study evaluated multiple ML algorithms to predict arsenic concentrations based on measured WQPs including pH, water temperature (WT), electrical conductivity (EC), salinity (WS), dissolved oxygen (DO), total dissolved solids (TDS), water hardness (WH), total

alkalinity (TA), chemical oxygen demand, chloride (CL), and biological oxygen demand. Descriptive statistics showed moderate variability and symmetrical distributions in WQPs, with arsenic level varying between 0.01 and 0.72 mg L⁻¹. Correlation analysis revealed positive associations between arsenic and all WQPs (WT, $r=0.29$ to TDS, $r=0.63$), except for a negative correlation with DO ($r=-0.41$). Among the models, MLP exhibited better predictive performance, achieving R^2 increments of 24.75%, 13.06%, 16.92%, and 5.15%, and RMSE reductions of 34.88%, 26.32%, 30%, and 12.5%, compared against MLR, RFR, SVR, and XGB algorithms, in that order. According to the finding, model performance was ranked as MLP > XGB > RFR > SVR > MLR. Notably, input combination 6 (TDS, Cl, EC, WH, TA, WS) enabled the MLP model to explain over 78.70% of the total variance and contributed to 91.83% of the total prediction accuracy. This research may offer important insights for assessing arsenic contamination in groundwater across Bangladesh's southeastern coastal area.

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Introduction

Groundwater arsenic contamination is a critical environmental and public health concern, particularly in Bangladesh's coastal zone (Huq et al., 2020). This area is part of the GBM (Ganges–Brahmaputra–Meghna) delta and naturally contains arsenic-bearing minerals that can enter groundwater aquifers under specific geochemical conditions such as reductive dissolution, microbial activity, and fluctuating redox states (Ahmad et al., 2018; Edmunds et al., 2015; Li et al., 2026). Groundwater arsenic levels in many parts of this coastal zone exceed the WHO (World Health Organization) recommended limit of 0.01 mg L^{-1} , posing serious health risks (Kumar & Pati, 2022). Prolonged exposure may cause skin lesions, internal cancers, cardiovascular disorders, developmental problems in children, and neurological impairments (Das, 2020; Edmunds et al., 2015). The crisis is further intensified by the community's heavy reliance on arsenic-prone shallow tube wells (Md Towfiqul Islam et al., 2020). Moreover, contaminated groundwater used for irrigation further causes bioaccumulation of arsenic in crops (Liu et al., 2025), thereby broadening exposure pathways (Rahaman et al., 2022). The spatial heterogeneity of arsenic distribution, along with the influence of surrounding hydrochemical conditions also complicates its management (Islam & Mostafa, 2023; Islam et al., 2023; Jeelani et al., 2020; Liang et al., 2018; Ullah et al., 2023). Consequently, examining associations linking arsenic levels with WQPs (water quality parameters) could prove essential for groundwater resource management.

A number of studies have investigated the complex interrelationship between arsenic concentration and various WQPs that influence its mobilization and persistence in groundwater (Chakraborty et al., 2015; Hassan & Westerhoff, 2024; Zhao et al., 2024). For instance, pH exerts a significant influence over adsorption and desorption of arsenic on mineral surfaces (Poursaeid, 2025). Under neutral to slightly alkaline conditions, arsenic tends to remain in the dissolved phase, increasing its mobility (Eberle et al., 2020; Winkel et al., 2012). Elevated WT (water temperature) may enhance microbial activity and redox reactions, affecting arsenic mobilization from aquifer sediments (Müller et al., 2022). High TDS (total dissolved solids), WS (water salinity), and EC (electrical

conductivity) can cause dissolution of arsenic-containing minerals (Aladejana et al., 2021; Uddin et al., 2024). Low DO (dissolved oxygen) typically favors reductive breakdown of iron and manganese oxides, releasing adsorbed arsenic (Donselaar et al., 2017; Erickson et al., 2021; Huq et al., 2020; Islam & Mostafa, 2023). WH (water hardness) and TA (total alkalinity) modify buffering capacity and affect redox potential, indirectly impacting arsenic solubility (Janjaroen et al., 2017). Elevated BOD (biochemical oxygen demand) and COD (chemical oxygen demand) levels promote microbial reduction processes under anaerobic conditions. These processes accelerate arsenic mobilization by transforming arsenic species and dissolving iron-bound minerals that typically immobilize arsenic (Bertin & Lett, 2009; N. Kumar et al., 2016). Additionally, chloride (Cl) can act as a competitive ion, influencing arsenic desorption from aquifer sediments (Fakhreddine et al., 2021; Morelli et al., 2018). Collectively, these parameters serve as indicators and drivers of arsenic behavior, and their integrated analysis may provide a new approach to model arsenic concentration. This study addresses this gap by focusing on hydrochemical WQPs to predict arsenic concentrations using advanced regression-based ML (machine learning) models in the southeastern coastal region of Bangladesh.

Several statistical approaches have been developed and adopted for predicting arsenic pollution in groundwater across different regions. Roy et al., (2013) applied a multivariate regression approach using only three WQPs, including Fe (iron), surface water level, and tube well age. Similarly, Amini et al., (2008) developed global arsenic probability maps using surface characteristics including soil composition, geological properties, climatic factors, and topographic elevation, through regression techniques and ANFIS (adaptive neuro-fuzzy inference system). Their model demonstrated overall good predictive performance during the testing phase. ANFIS explained 65% of variability for both reducing environments and high pH/oxidizing regions, compared to 59% and 56% using linear regression, respectively. Although these models demonstrated statistical significance, they lacked comprehensive hydrochemical variables that could better explain arsenic mobilization. LR (logistic regression) and MLOR (multivariate logistic regression) have also been used (Zhao et al., 2024); however, these models often assume

linear relationships and may underperform with complex, non-linear groundwater datasets. For instance, Zhao et al., (2024) reported an MLOR prediction accuracy of 72.18% in Bangladesh, indicating limited generalizability. Tan et al., (2020) and Chakraborty et al., (2020) also found LR-based models less effective when hydrochemical predictors were involved.

Given these limitations, recent studies increasingly use ML (machine learning) algorithms to predict arsenic concentrations due to their ability to process large, high-dimensional, and non-linear datasets of water quality (Basak et al., 2025; Hussain et al., 2025). ML models are especially valuable for rapid, cost-effective arsenic monitoring in remote or resource-limited regions where frequent sampling and laboratory analyses are costly and impractical (Cho et al., 2011; García et al., 2023; Ji et al., 2024). For example, Zhou et al., (2024) applied RFR (random forest regression) and RFC (random forest classifier) to arsenic prediction in the Hetao Basin and Bangladesh, with RFR explaining up to 80% of variability and RFC achieving over 91% validation accuracy. However, regional heterogeneity remains a challenge for model generalizability. Akajiaku et al., (2025) predicted iron concentrations in groundwater using MLR and RFR, with MLR outperforming RFR ($R^2=0.92$, $MAE=0.13 \text{ mg L}^{-1}$, $RMSE=0.15 \text{ mg L}^{-1}$). However, the study was limited by the small dataset (50 wells) and the lack of additional hydrogeological variables, which may reduce model accuracy. Kumar and Pati (2022) applied several tree-based classifiers and reported OFM (optimized forest model) was most accurate with 80.64% classification accuracy. Ibrahim et al., (2022) evaluated advanced ML models like XGB (extreme gradient boosting), LGB (light gradient boosting), and GRNN (generalized regression neural networks), reporting high performance scores ($R^2>0.93$). However, large set of input variables in their study may increase model complexity, as the study did not evaluate different input combinations to select the most relevant variables. Bindal and Singh (2019) used a hybrid RFR with univariate feature selection to predict arsenic in Uttar Pradesh, though the reliance on proxy variables rather than direct WQPs limited its broader applicability. Cao et al., (2018) showed that a hybrid PC-ANN (principal component-artificial neural network) model outperformed traditional methods. Overall, most studies have applied ML classifiers to categorize arsenic

concentrations as “high,” “moderate,” or “low” based on thresholds such as the WHO standard of 0.01 mg L^{-1} , using physical and spatial predictors like soil chemistry, elevation, and land cover (Kumar & Pati, 2022; Zhao et al., 2024). While useful for identifying contamination zones, these classification models provide limited insight into actual concentration levels, which are crucial for arsenic monitoring. Moreover, regression-based ML models have been rarely applied to predict arsenic concentrations with high accuracy, particularly in regions such as the southeastern coastal area of Bangladesh. Furthermore, the potential of hydrochemical variables as predictors remains unexplored in arsenic concentration modeling.

Therefore, this study aims to utilize advanced ML algorithms to predict arsenic levels in groundwater based on key WQPs in the Bangladesh’s southeastern coastal area. The specific objectives are threefold: (1) to determine the critical WQPs which significantly affect arsenic concentration in groundwater from coastal zone; (2) to evaluate the predictive capability of models developed with different ML techniques, including MLR (multiple linear regression), RFR, SVR (support vector regression), XGB, as well as MLP (multilayer perceptron) for arsenic concentration prediction; and (3) to identify optimal ICs (input combinations) for ML models with the most influential WQPs that significantly affect the arsenic concentrations.

This study may provide several key innovations and contributions to arsenic prediction research by focusing on hydrochemical WQPs as predictors. Unlike previous studies that relied on physical, geological, or spatial proxies (Amini et al., 2008; Roy et al., 2013; Zhao et al., 2024), it integrates a comprehensive set of hydrochemical variables to better understand the arsenic mobilization. The application of multiple regression-based ML models addresses the common limitation of classification based approaches (Kumar & Pati, 2022; Zhou et al., 2024), which provide only threshold values rather than actual arsenic concentrations. Moreover, optimized input combinations of WQPs enhance model performance and reduce complexity of the models. By focusing on the highly arsenic-contaminated coastal region of Bangladesh (Ayers et al., 2016), the study may fill an important regional knowledge gap and provide insights for improved groundwater management.

Materials and methods

Study area

The study was conducted within three southeastern coastal districts in Bangladesh, Noakhali, Feni, and Lakshmipur, located along the Bay of Bengal's northern coast in the Chittagong Division (Fig. 1). The landscape comprises low-lying deltaic, estuarine plains and tidal flats formed by sediment deposition from large river systems. These districts were chosen as representative coastal zones facing severe groundwater contamination challenges in Bangladesh. Due to the limited availability of reliable surface water, groundwater serves as the primary source for drinking and for irrigation (Dipty et al., 2023; Islam et al., 2021; Sultana & Farhana, 2024). Noakhali is highly vulnerable to groundwater issues, including saline intrusion, shallow aquifers, and over-extraction, affecting millions of residents. The situation is worsened by tidal surges, riverbank erosion, and reduced aquifer recharge (Kamran et al., 2024). Feni, located to the northeast, shares a similar hydrological setting but is increasingly affected by agricultural runoff and urbanization, which degrade groundwater quality. Westward, Lakshmipur experiences seasonal waterlogging, salinity, and contamination from domestic and agricultural sources (Chakraborti et al., 2010; Hossain, 2006). All three districts have extensive shallow tube well use, but lack adequate monitoring infrastructure, heightening the risk of arsenic contamination.

Sampling design

Groundwater samples were obtained exclusively throughout the dry season (November–March) to reduce the influence of monsoonal recharge and to capture the baseline hydrochemical conditions crucial for understanding arsenic behavior in aquifers (Haque et al., 2016; Stopelli et al., 2020). In total, 600 groundwater samples were obtained from shallow tube wells (exceeding 100 feet depth) across the coastal districts of Noakhali, Feni, and Lakshmipur, ensuring comprehensive spatial coverage and hydrological representation. Sampling sites were initially selected using topographic maps, hydrological characteristics, population density, and potential arsenic risk zones (Navoni et al., 2014). Satellite imagery

from Google Earth Pro and Sentinel-2 was used to identify suitable locations; GIS-based spatial overlays of land use and water accessibility were applied to finalize the sampling points (Fig. 1). Field visits were conducted prior to sampling to verify local conditions and confirm site suitability. At each selected location, groundwater was extracted using hand pumps after flushing for several minutes to obtain fresh aquifer water. Samples were collected in 500 mL polyethylene bottles that had been thoroughly cleaned with 10% nitric acid (HNO_3) and rinsed with ultrapure water. Each bottle was triple rinsed with the respective well water before final collection to avoid contamination. Immediately after collection, the bottles were tightly sealed and stored in ice-packed coolers to maintain sample integrity. All samples were transferred to Department of Environmental Science and Disaster Management laboratory at Noakhali Science and Technology University, where they were analyzed for selected physicochemical parameters and arsenic concentration using standard procedures. Proper quality assurance and sample handling protocols were maintained throughout the sampling process to ensure accuracy.

Measurement of physiochemical parameters

Water physicochemical parameters, such as BOD, COD, Cl (chloride), DO (dissolved oxygen), EC (electrical conductivity), pH, TA (total alkalinity), TDS, WH (total hardness), WT (water temperature), and WS (water salinity) were analyzed to evaluate groundwater quality and arsenic contamination. While several earlier studies have incorporated geological, climatic, or spatial predictors (Amini et al., 2008; Zhao et al., 2024), WQPs were prioritized here due to their direct influence on arsenic mobilization, as they are known to control the solubility and transport of arsenic (Chakraborty et al., 2015; Cho et al., 2011; Hassan & Westerhoff, 2024). Limiting the study to WQPs allowed for a more targeted assessment of water-chemistry influences on arsenic contamination and facilitated consistent data collection across all sampling sites. Standardized and validated analytical methods were adopted to ensure the accuracy, reproducibility and reliability of results across all sampling sites (Kothari et al., 2021). Field-sensitive parameters such as pH, EC, WS, DO, TDS and WT were analyzed in situ using pre-calibrated multi-parameter

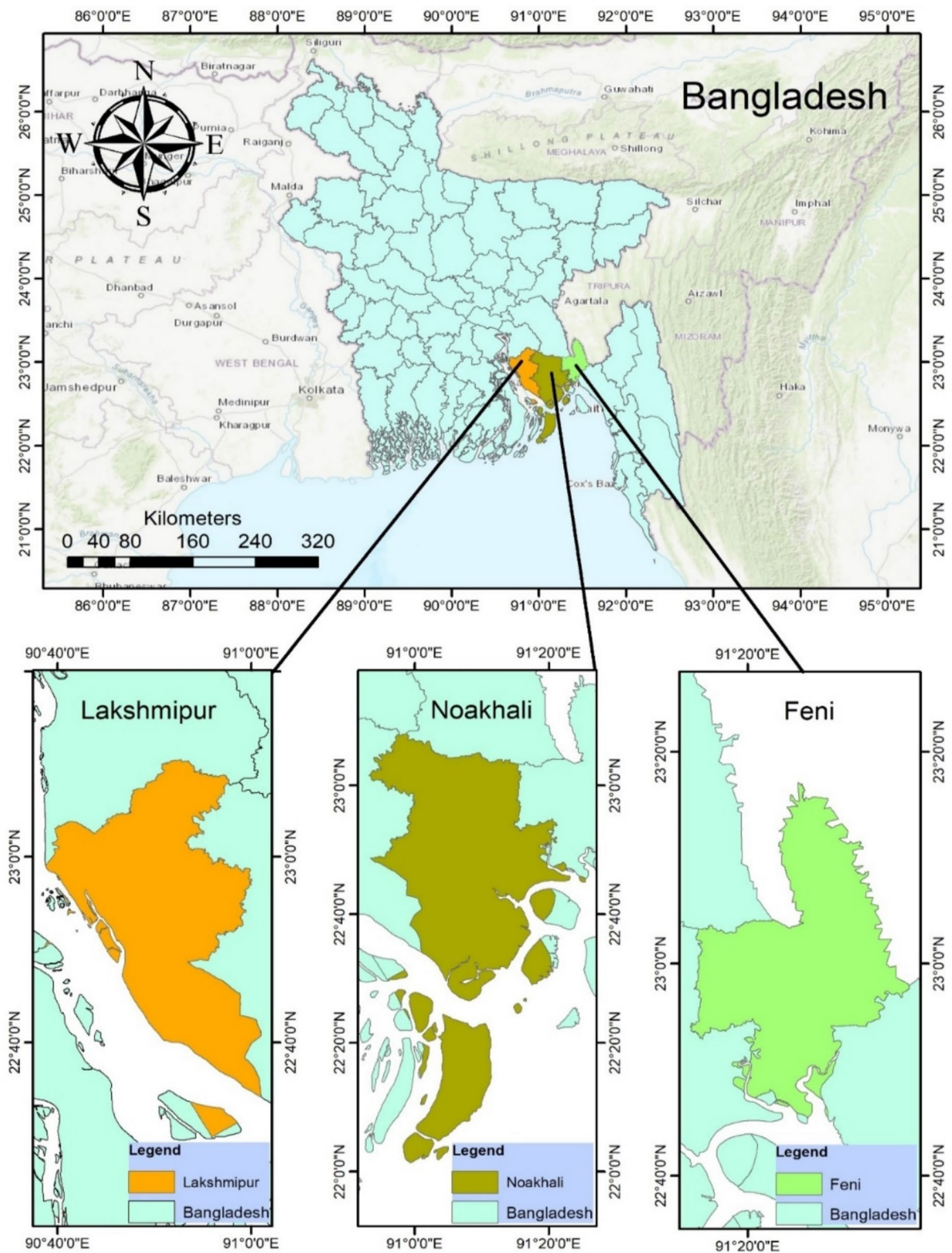


Fig. 1 Map showing the three southeastern coastal regions of Bangladesh

inquiries listed in Table 1. For the measurement of WH, BOD, COD, Cl, and TA samples were collected, well-maintained, and analyzed in accordance to established standard protocols as outlined in Table 1. Comprehensive quality control and quality assurance procedures were followed, including the use of ultrapure deionized water (18.2 MΩ-cm) generated by a Barnstead Easypure-II RF unit (Model D7031, Thermo Scientific, USA), routine calibration of instruments before each sampling session, and rigorous cleaning of all glassware through detergent soaking, acid washing with 10% HNO₃, and repeated rinsing with ultrapure deionized water to prevent cross-contamination.

Determination of arsenic concentration

To determine arsenic concentrations in groundwater samples, high-purity analytical-grade reagents were used to ensure accuracy and minimize contamination. Concentrated nitric acid (HNO₃, 65% w/v) and hydrochloric acid (HCl, 37% w/v) were procured from Merck (Darmstadt, Germany). Ultrapure water (18.2 MΩ-cm resistivity), generated using a Barnstead Easypure-II RF system (Model D7031, Thermo Scientific, USA), was used throughout all analytical steps to avoid the external impurities. Certified reference materials (CRMs) for elemental calibration were obtained from Sigma-Aldrich (St. Louis, Missouri, USA) to verify instrumental performance and ensure traceability of the measurements. Groundwater samples were digested using the aqua regia to break down dissolved and particulate-bound arsenic.

For each sample, 100 mL of groundwater was mixed with 2 mL of concentrated HNO₃ and 5 mL of HCl, followed by controlled heating on a hot plate at 90–95 °C until the volume was reduced to approximately 50 mL. This digestion approach facilitates the release of arsenic from complex matrices. After cooling, the digests were filtered through pre-cleaned Whatman filter paper and quantitatively diluted to a final volume of 100 mL using ultrapure water. All digestion procedures were performed inside a certified fume hood to maintain laboratory safety and prevent acid vapor exposure. The prepared digested samples were subsequently analyzed using an air-acetylene flame Atomic Absorption Spectrophotometer (AAS) (Model AA240FS, Varian, Australia). Instrument calibration and blank determinations were performed prior to analysis to maintain data quality. Comprehensive quality assurance and quality control (QA/QC) measures along with the key analytical performance parameters are summarized in Table 2.

Data preprocessing

The successful prediction of groundwater arsenic contamination relies on identification of the most relevant input variables for ML models. To achieve this, linear relationships between predictor variables and arsenic concentrations were examined through Pearson correlation coefficient analysis among arsenic levels alongside many WQPs. Based on the strength of the correlation coefficients with arsenic, input variables were ranked and grouped to formulate various ICs (input combinations) for analysis (Basak et al.,

Table 1 Measurement methods for physiochemical parameters

Parameter	Unit	Method	References
pH		Multi-Parameter, Model: HI98194, Hanna Instruments	Anand et al. (2024)
EC	μS cm ⁻¹	Multi-Parameter, Model: HI98194, Hanna Instruments	Singh et al. (2024)
WS	PSU	Multi-Parameter, Model: HI98194, Hanna Instruments	Singh et al. (2024)
DO	mg L ⁻¹	Multi-Parameter, Model: HI98194, Hanna Instruments	Anand et al. (2024)
TDS	mg L ⁻¹	Multi-Parameter, Model: HI98194, Hanna Instruments	Anand et al. (2024)
WT	°C	Thermometer, Model: HI98501, Hanna Instruments	Mansour et al. (2024)
WH		Titration (Double burette method)	Dhoke (2024)
BOD	mg L ⁻¹	5-day incubation method	Mckenzie (2003)
COD	mg L ⁻¹	Dichromate digestion (AWWA 5220-C)	Ching and Redzwan (2017)
Cl ⁻	mg L ⁻¹	Mohr's titration	Shukla and Arya (2018)
TA	mg L ⁻¹	Neutralization titration	Dhoke (2023)

Table 2 Quality control, quality assurance and analytical parameters for arsenic determination in groundwater samples using flame atomic absorption spectrophotometry

Parameter	Value
Certified value ($\mu\text{g L}^{-1}$)	43.93
Observed value ($\mu\text{g L}^{-1}$)	42.05
Recovery (%)	95.7
Method Detection Limit (MDL) ($\mu\text{g L}^{-1}$)	<0.5
Relative Standard Deviation (RSD) (%)	1.04
Hollow Cathode Lamp Current (mA)	10
Wavelength (nm)	193.7
Slit Width (nm)	0.5
Limit of Detection (LOD) ($\mu\text{g L}^{-1}$)	0.52
Limit of Quantification (LOQ) ($\mu\text{g L}^{-1}$)	1.5
Calibration Range ($\mu\text{g L}^{-1}$)	10 – 1000
Measurement Uncertainty ($\pm$) (K=2)	16%

2022). Variable exhibiting the highest correlation coefficients was introduced to the models initially, and subsequent variables were added sequentially to generate ICs and assess improvements in ML model predictive accuracy with these ICs. This method helps identify the optimum combination of WQPs that effectively contributes to arsenic prediction whereas reducing the number of input parameters, thus enhancing the model's simplicity and practical use. For ensuring data quality and reliability, boxplot evaluation was employed to identify outliers. No outliers were found within the dataset. Following this, the Z-score normalization technique was applied to standardize input variables, scaling all data values to a uniform range, which helps enhance model training and ensures that each variable contributes equally to the learning process. The standardization was performed utilizing the following Eq. (1).

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

In Eq. (1), Z denotes the standardized value of a variable, x and μ indicate the individual observation and represents the dataset mean for all values for that variable, and σ refers the standard deviation.

Machine learning algorithms

This study utilized five ML algorithms: MLR (multiple linear regression), RFR (random forest

regression), SVR (support vector regression), XGB (extreme gradient boosting), as well as MLP (multi-layer perceptron). MLR was selected as the baseline to compare the performances of other models. RFR, SVR, and XGB were selected for their proven ability to handle non-linear relationships to provide accurate predictions, particularly in environmental and groundwater studies (Akajiaku et al., 2025; Ibrahim et al., 2022; S. Kumar & Pati, 2022; Zhao et al., 2024). The MLP, a deep learning neural network model, was also used for evaluating complex, non-linear interactions among WQPs. Although MLP has been rarely applied in groundwater arsenic prediction, its architecture may be useful to model intricate parameter interactions. These models were preferred over alternatives such as CatBoost, LSTM, or GBM because they have demonstrated superior performance in similar arsenic and groundwater contamination prediction studies while remaining computationally efficient and widely validated (Akajiaku et al., 2025; Akhlaq et al., 2024; Chen et al., 2026; Ibrahim et al., 2022; Zhao et al., 2024). Collectively, the selected models present a comprehensive framework for predicting arsenic contamination. The next section provides an overview discussion of these five particular ML techniques.

Multiple linear regression (MLR)

MLR represents a traditional statistical approach for examining relationships among one response variable and two or additional predictor variables. MLR assumes linearity between inputs and outputs and is commonly applied owing to its straightforward implementation and ease of interpretation (Fig. 2a). This approach serves as a baseline model for the purpose of comparison with more complex algorithms (Basak et al., 2022). The general MLR model defined as Eq. (2).

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n + \varepsilon \quad (2)$$

where Y_i denotes the estimated concentration of arsenic, β_1 signifies the intercept term, $\beta_1, \beta_2, \beta_3 \dots \beta_n$ correspond to coefficients associated with each predictor $X_1, X_2, X_3, \dots X_n$, while ε denotes the residual error term.

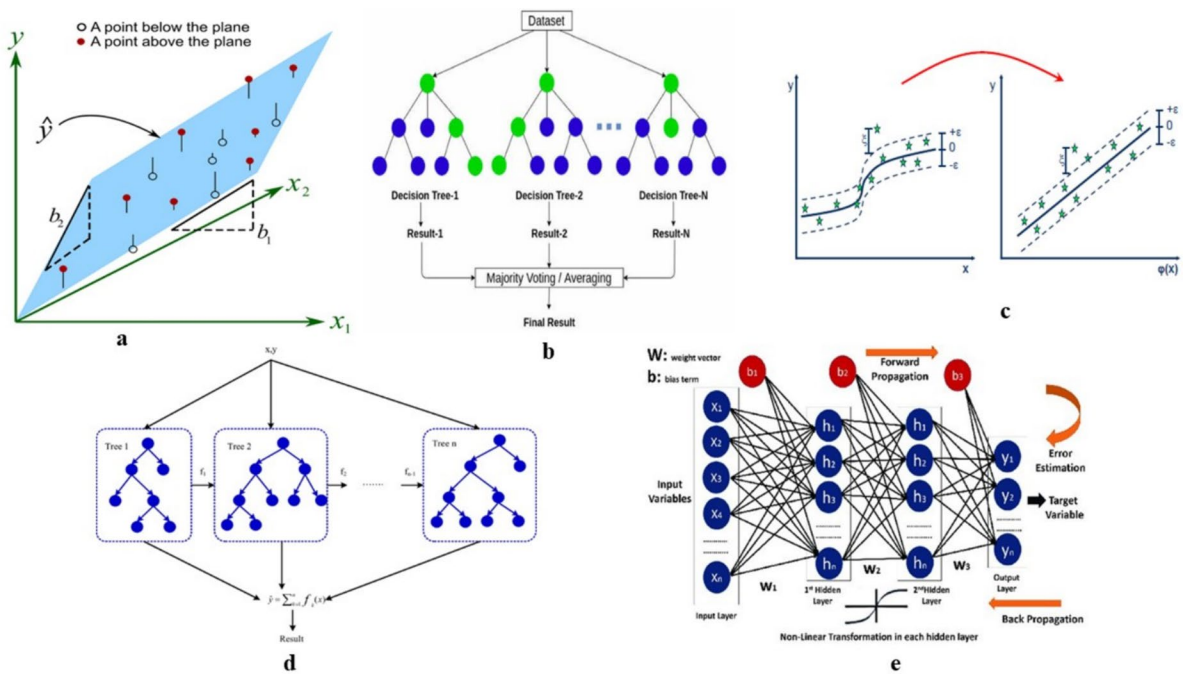


Fig. 2 Illustrations architecture for **a** MLR, **b** RFR, **c** SVR, **d** XGB, and **e** MLP models

Random forest regression (RFR)

RFR represents a powerful ensemble-based method that constructs numerous decision trees throughout training and aggregates their predictions through averaging for enhanced precision and reducing the likelihood of overfitting (Breiman, 2001) (Fig. 2b). During training, the algorithm utilizes the Gini Impurity Index to select the optimal splits in each decision tree. A node's impurity can be measured using the Gini Index, guiding the model to create the most informative divisions (Sahani & Ghosh, 2021). The formula for calculating the Gini Index is provided in Eq. (3).

$$\text{Gini}(X_i) = \sum_{j=1}^J P(X_i = L_j)(1 - P(X_i = L_j)) = 1 - \sum_{j=1}^J P(X_i = L_j)^2 \quad (3)$$

where $P(X_i = L_j)$ denotes the probability that observation X_i belongs to class L_j , and J represents the over-all number of possible classes (with $j \geq 2$).

Averaging the outputs from all individual decision trees within the ensemble yields the final

prediction for the RFR model. This is expressed mathematically in Eq. (4).

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N F_i(x) \quad (4)$$

where $F_i(x)$ denotes the decision tree's ensembles, N denotes the number of trees, and $\hat{y}$ represent average ensemble projection.

Support vector regression (SVR)

The SVR approach constructs a linear regression model within a specified boundary after converting input variables into a higher-dimensional feature

space (Fig. 2c). It utilizes an ε -insensitive loss function, allowing deviations within a predefined ε threshold to be disregarded. By applying a kernel function, SVR effectively captures nonlinear relationships through reliable feature transformation (Amani et al.,

2023). The primary goal is to determine a mathematical function capable of estimating a response outcome from the predictor features, minimizing discrepancies between predicted output and its corresponding actual value y within an acceptable tolerance threshold defined by ε . The fundamental equation governing the SVR model is presented in Eq. (5), while Eq. (6) specifies the equation for the ε -insensitive tube, following subject to the limitations outlined in Eq. (7). The SVR regression function is expressed as follows:

$$y = f(x) = \omega \cdot \phi(X) + b \quad (5)$$

In Eq. (5) ω indicates the coefficient vector, $\phi(x)$ denotes the transformation of input data x into a higher-dimensional feature space, while b represents the intercept term. The SVR optimization problem minimizes both the model complexity and the prediction error outside the epsilon margin, as formulated below:

$$\min_{\omega, b, \xi, \xi^*} \frac{1}{2} \|\omega\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (6)$$

Subject to constraints:

$$\begin{cases} y_i - f(x_i) \leq \varepsilon + \xi_i \\ f(x_i) - y_i \leq \varepsilon + \xi_i^* \\ \xi_i, \xi_i^* \geq 0 \end{cases} \quad (7)$$

where ξ_i and ξ_i^* are loose variables indicating the allowed prediction error in the model, C represents the penalty parameter that controls trade-offs between model complexity and tolerance, while ε specifies the width of the error-insensitive tube.

Extreme gradient boosting (XGB)

XGB represents an advanced ML algorithm built on a gradient boosting framework. This method constructs an ensemble of decision trees sequentially, with each new tree aimed at correcting residual errors produced by made by preceding trees (Fig. 2d). To enhance model interpretability, improve computational efficiency, prevent overfitting, XGB incorporates optimization techniques such as regularization and pruning (Chen & Guestrin, 2016). The model updates its prediction at each iteration t is updated through addition of a newly constructed tree to the existing model. This is expressed as Eq. (8).

$$f_i^{(t)} = \sum_{k=1}^t f_k(x_i) = f_i^{(t-1)} + f_t(x_i) \quad (8)$$

where $f_i^{(t)}$ denotes the model prediction at iteration t , while $f_i^{(t-1)}$ represents the prediction from the previous iteration. The term $f_t(x_i)$ refers to the tree added at iteration t , and x_i denote the input feature vector.

$$Obj^{(t)} = \sum_{k=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^t \Omega(f_i) \quad (9)$$

In the equation, l denotes the loss function, which measures discrepancies between forecasted values and measured observations, $\hat{y}_i$ is obtained by summing the cumulative output from all residual trees, n denotes the actual observed values, while $\Omega(f_i)$ refers to the penalty term, as specified in Eq. (10).

$$\Omega(f) = \gamma + \frac{1}{2} \lambda \|\omega\|^2 \quad (10)$$

The regularization term consists of several components, including T (leave number), ω (weight vector), λ (regularization coefficient), and γ (the lowest loss parameter).

Multilayer perceptron (MLP)

MLP constitutes an ANN (artificial neural network) type architecture that comprises an input layer, one or more hidden layers, and an output layer (Fig. 2e). Neurons within each layer establish connections to neurons in the subsequent layer through regulated connections, which determine the intensity of the signal passed (Arulmozhi et al., 2021). These weights are adjusted during training, enabling the network to capture complex relationships between input features and target values. During training, MLP uses the backpropagation algorithm, which iteratively updates the weights to minimize the prediction error. This occurs by measuring gradients of the error relative to network weights and adjusting them in the direction that reduces the error. A three-layer MLP can be statistically represented as Eq. (11) (Deb et al., 2025).

$$y_p = f_0 \left[\sum_{j=1}^n W_{kj} f_h \left(\sum_{j=1}^m W_{ji} X_i + W_{jb} \right) + W_{kb} \right] \quad (11)$$

where y_p represents the model output, f_0 represent the activation function in the final layer, and f_h is the hidden layer's activation function. The input variable is denoted by x_i , w_{ji} is the connection weights between

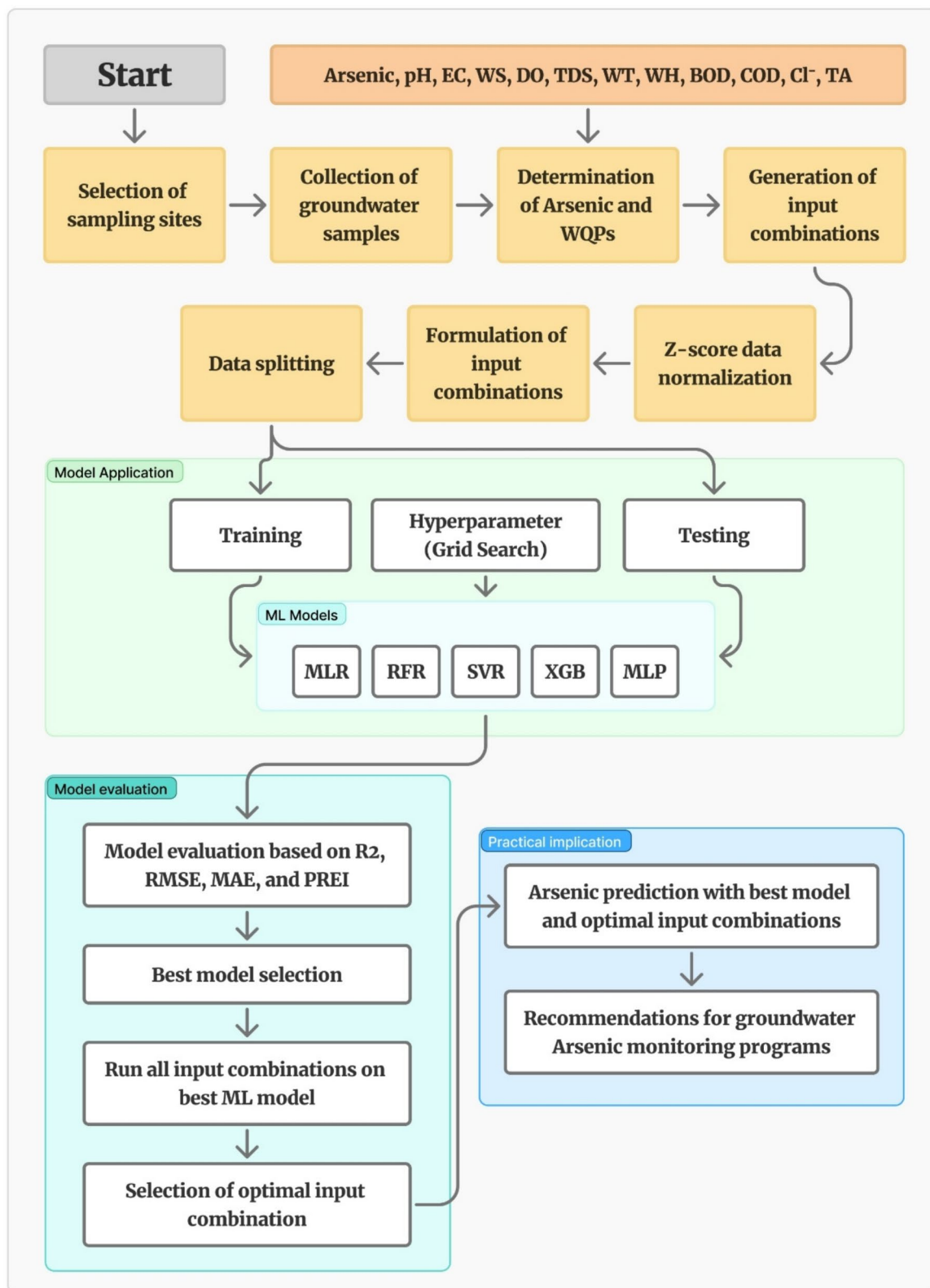


Fig. 3 Comprehensive flowchart for groundwater arsenic prediction model

the input and hidden layers, w_{kj} represent the connection weight linking the hidden and output layers. The bias terms w_{ji} and w_{kb} represent the hidden and output layers, respectively. Additionally, m and n indicate the number of neurons in the input and hidden layers, respectively.

$$W_x^* = W_x - a \left(\frac{\partial \text{Error}}{\partial W_x} \right) \quad (12)$$

where, W_x^* denotes the modified weight and W_x represent the previous weight value, a denote the term learning rate and the term $\frac{\partial \text{Error}}{\partial W_x}$ denotes signifies the gradient of the error function about weight. The goal of this method is to minimize the output prediction error.

$$E = \sum_{p=1}^P E_p = \sum_{p=1}^P \cdot \sum_{k=1}^n (y_p - y_a)^2 \quad (13)$$

In Eq. (13), E signifies the overall cumulative error, E_p denotes the individual squared error corresponding to the p^{th} training instance, and P refers to the total number of training samples.

Model implementation and evaluation

For this study, Anaconda's with Jupyter Notebook interface was used for implementing five ML models: MLR, RFR, SVR, XGB, and MLP. Required modules from sklearn were utilized to support the execution of these models within Jupyter Notebook. For identifying an optimal data split, multiple training and testing splits (70:30, 80:20, and 90:10) were tested. The 80:20 split was selected after comparing multiple ratios, as it consistently provided the best predictive accuracy across the ML models. Additionally, a five-fold cross-validation approach was applied to minimize overfitting and provide a more reliable estimate of the models' ability to generalize to unseen data. The mean performance across the five-fold cross-validation was considered as the training accuracy. Furthermore, the remaining 20% of the original data was kept for independent assessment of the predictive performance. Hyperparameter optimization was executed with the grid search procedure. For the RFR, the parameters subjected to tuning

comprised the number of estimators (range 10–100), the minimum sample size within terminal nodes (range 1–30), the number of predictors evaluated per split (range 1–5), and the maximum allowable depth of the trees (range 1–100). In the case of the SVR, the model was configured with a radial basis function kernel, penalty parameter C values drawn from a uniform discrete random distribution (Ud) within the interval [1, 100], a fixed gamma parameter of 1, and an epsilon value of 0.1. For the XGB algorithm, hyperparameter refinement included minimum child weight (range 1–5), gamma (range 0.1–1), subsampling ratio (range 0.5–1.0), a learning rate of 0.1, and maximum tree depth (range 3–5). For the MLP, the optimized hyperparameters consisted of hidden layer dimension (3), activation function (logistic), regularization coefficient (0.0001), learning rate strategy (constant), and optimization solver (lbfgs). These systematic adjustments led to notable improvements in training performance across all implemented ML models. Three statistical indicators were employed to evaluate model performance: including the R^2 (coefficient of determination) (Eq. (14)), MAE (mean absolute error) (Eq. (15)), and RMSE (root mean square error) (Eq. (16)). A higher R^2 value, combined with lower MAE and RMSE, values, reflects improved model performance (Basak et al., 2020; Bui et al., 2020). Furthermore, PREI (percentage of relative error index) (Eq. (17)) was used to evaluate uncertainty in ML model predictions of arsenic contamination (Sajib et al., 2023). When PREI values are positive, this shows that models underestimated the arsenic contamination, when negative PREI values denote that models overestimated them (Bui et al., 2020). To evaluate the distribution and reliability of the predicted arsenic contamination values, various analytical plots were executed, including actual versus predicted line and scatter plots, cumulative distribution plots of residuals, and Taylor diagrams for each model utilizing the R programming environment (Version 4.4.2). A comprehensive flowchart of the data collection, data preprocessing, model execution, and performance evaluation is presented in Fig. 3.

$$R^2 = 1 - \frac{\sum_{i=1}^n (O_i - P_i)^2}{\sum_{i=1}^n (O_i - \bar{O})^2} \quad (14)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |O_i - P_i| \quad (15)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (O_i - P_i)^2} \quad (16)$$

$$PREI = \left(\frac{O_i - P_i}{O_i} \right) \times 100 \quad (17)$$

where n signifies the cumulative count of data points, while O_i and P_i correspond to actual and forecast values, respectively. The bar notation used to represent the mean value of variables.

Results

Descriptive statistics of WQPs and arsenic

Table 3 shows a detailed summary of the descriptive statistics for the measured WQPs and arsenic concentration within groundwater. Overall ranges of pH, WT, EC, WS, DO, TDS, WH, TA, COD, Cl, BOD, arsenic were 3.50–8.30, 19.14–26.22 °C, 200.89–2880 $\mu\text{S cm}^{-1}$, 0.17–2.31 PSU, 3.21–6.26 mg L^{-1} , 103.95–1380.30 mg L^{-1} , 68.88–323.09 mg L^{-1} , 166.32–1078.30 mg L^{-1} , 49.54–184.00 mg L^{-1} , 80.46–1491.95 mg L^{-1} , 0.97–4.83 mg L^{-1} , and 0.01–0.72 mg L^{-1} , respectively. The STDs (standard deviations) for these parameters reflected the degrees of variability across

the sampling sites. The highest STDs were observed for EC (317.29 $\mu\text{S cm}^{-1}$), Cl (175.45 mg L^{-1}), TDS (148.09 mg L^{-1}), TA (91.54 mg L^{-1}), WH (32.93 mg L^{-1}), and COD (15.24 mg L^{-1}). The quartile distribution (Q1, Q2, and Q3) for the WQPs demonstrates a relatively symmetrical pattern around the median, as the second quartile values closely align with the respective means. This alignment indicates moderate variability in water quality characteristics and suggests the absence of significant skewness in the data distribution.

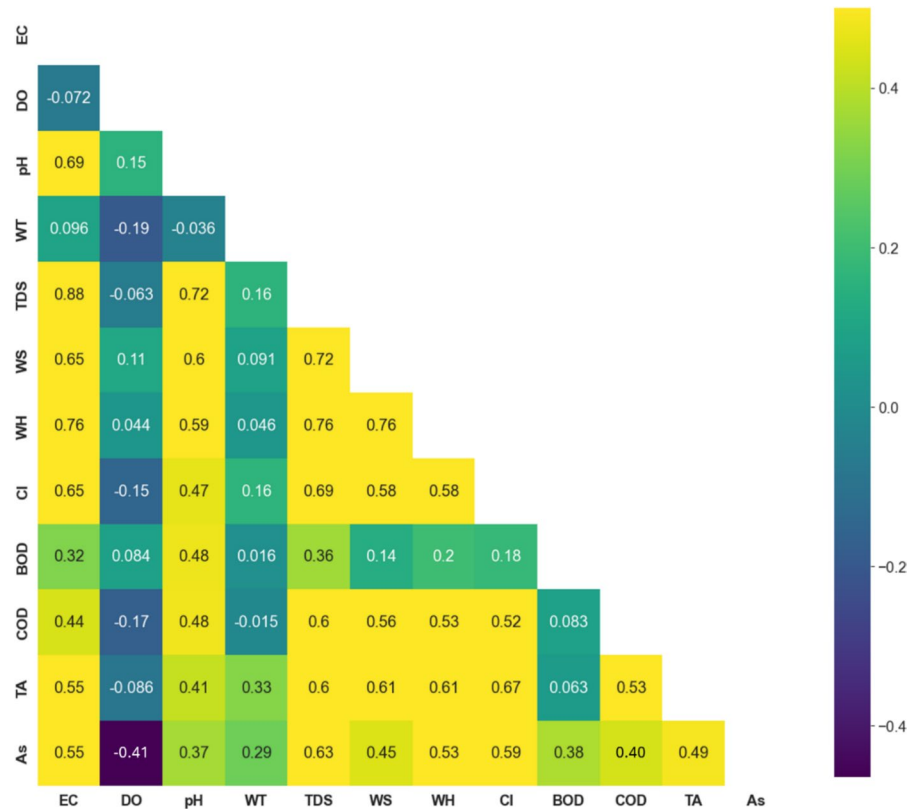
Correlation analysis between WQPs and arsenic

Figure 4 presents the Pearson correlation coefficients between arsenic and various WQPs, showing several strong and moderate relationships. Arsenic exhibits strong positive correlations with TDS ($r=0.63$), Cl ($r=0.59$), EC ($r=0.55$), WH ($r=0.53$), TA ($r=0.49$), WS ($r=0.45$), indicating that higher arsenic levels are associated with increased salinity and ionic content. Moderate positive correlations are also observed with COD ($r=0.40$) and BOD ($r=0.38$), while a moderate negative correlation is seen with DO ($r=-0.41$), indicating a potential link between lower oxygen levels and elevated arsenic concentrations. Among the WQPs, most correlations are strong and positive, except for DO, WT, and BOD. Very strong positive correlations ($r>0.75$) were observed between EC and TDS, EC and WH, and WH and WS. In contrast, DO, WT, and BOD exhibited very weak negative to weak positive correlations with other WQPs, ranging from -0.19 to 0.36. Based on correlation analysis,

Table 3 Summary statistics of WQPs and arsenic concentration

Parameter	Unit	Mean	Min	Q1	Q2	Q3	Max	STD
Arsenic	mg L^{-1}	0.31	0.01	0.27	0.31	0.35	0.72	0.07
BOD	mg L^{-1}	1.96	0.97	1.65	1.89	2.18	4.83	0.47
Cl	mg L^{-1}	853.91	80.46	758.29	861.59	962.84	1491.95	175.45
COD	mg L^{-1}	122.12	49.54	114.44	122.07	130.63	184	15.24
DO	mg L^{-1}	4.85	3.21	4.65	4.84	5.02	6.26	0.32
EC	$\mu\text{S cm}^{-1}$	1408.83	200.89	1220.14	1396.07	1600.48	2880	317.29
pH	–	6.75	3.50	6.38	6.84	7.20	8.30	0.69
TA	mg L^{-1}	568.81	166.32	524.13	566.39	614.77	1078.3	91.54
TDS	mg L^{-1}	730.05	103.95	645.62	728.32	820.94	1380.3	148.09
WH	mg L^{-1}	183.48	68.88	165.03	182.38	201.25	323.09	32.93
WS	PSU	0.96	0.17	0.82	0.95	1.09	2.31	0.23
WT	°C	21.68	19.14	21.2	21.66	22.13	26.22	0.8

Fig. 4 Pearson correlation analysis exhibiting associations among WQPs with arsenic concentration. Correlation strength is represented by color intensity



no WQPs exceeded the multicollinearity threshold of $|r| > 0.90$. Therefore, multicollinearity is unlikely to have significantly affected the machine learning models (MLR, RFR, SVR, MLP) used in this study.

Performance evaluation of ML models

A comparative analysis of the MLR, RFR, SVR, XGB, and MLP algorithms was conducted to assess their predictive performance for arsenic concentration in groundwater based on the selected WQPs. The results demonstrated that all algorithms performed reasonably well in both training and testing phases (Table 4).

For the training phase, R^2 , RMSE, and MAE values ranged from 0.735 to 0.941, 0.017 to 0.038 mg L^{-1} , and 0.016 to 0.035 mg L^{-1} , respectively, while in the testing phase, the respective values ranged from 0.687 to 0.857, 0.028 to 0.043 mg L^{-1} , and 0.026 to 0.039 mg L^{-1} . Among evaluated algorithms, MLP showed superior accuracy with maximum R^2 alongside minimum RMSE and MAE, whereas MLR demonstrated the lowest performance with minimum R^2 combined with the maximum error values in both training and testing phases. Notable improvements over MLR, RFR, SVR, and XGB algorithms were exhibited by the MLP model. In training phrase, R^2

Table 4 Performance evaluation metrics of the ML models during the training and testing stages. R^2 reflects predictive accuracy of the algorithms, while RMSE and MAE represent prediction error

Model	Training			Testing		
	R^2	RMSE	MAE	R^2	RMSE	MAE
MLR	0.735	0.038	0.035	0.687	0.043	0.039
RFR	0.868	0.027	0.025	0.758	0.038	0.034
SVR	0.824	0.031	0.028	0.733	0.040	0.036
XGB	0.882	0.025	0.023	0.815	0.032	0.029
MLP	0.941	0.017	0.016	0.857	0.028	0.026

values increased by 28.03%, 8.41%, 14.20%, and 6.69%, while RMSE values decreased by 55.26%, 37.04%, 21.28%, and 32%, respectively, for MLP relative to MLR, RFR, SVR, and XGB. Similarly, in the testing stage, R^2 values increased by 24.75%, 13.06%, 16.92%, and 5.15%, while RMSE values decreased by 34.88%, 26.32%, 30%, and 12.5%, respectively, for MLP relative to MLR, RFR, SVR, and XGB. Overall, the models ranked in the following order based on their testing phase performance: MLP > XGB > RFR > SVR > MLR.

Figure 5 shows the line and scatter plots displaying relationships between measured versus predicted arsenic concentrations during the testing phase. The line graphs demonstrate that all models closely fitted the measured data, indicating high prediction accuracy. In the scatter plots, predictions from the MLP algorithm aligned most closely with the 1:1 reference line, outperforming MLR, RFR, SVR, and XGB algorithms. Figure 6 shows uncertainty evaluation through PREI metrics, which remained within $\pm 20\%$ across all models. These outcomes emphasize the reliability of the ML models, especially MLP, in predicting arsenic concentrations with minimal uncertainty. Furthermore, cumulative distribution analysis (Fig. 7), which contrasts the observed versus predicted outputs during the testing stage, offers a more comprehensive assessment of model effectiveness. The MLP demonstrated superior precision, with 96.67% of residuals constrained within the interval of -0.05 to 0.05 mg L^{-1} . In comparison, the corresponding proportions for the MLR, RFR, SVR, and XGB models were 76.67%, 85.00%, 82.50%, and 91.67%, respectively. The Taylor diagram (Fig. 8) further corroborates these findings, revealing that the MLP achieved the minimum RMSE alongside the highest correlation coefficient, positioning it nearest to the reference marker and thereby signifying its superior predictive capability for arsenic concentration.

Prediction of arsenic concentration using optimal input combination

Based on the correlation analysis between WQPs and arsenic concentrations presented in Fig. 4, eleven ICs (input combinations) of WQPs were sequentially constructed and employed for training and testing MLR along with MLP models for predicting arsenic concentration. TDS exhibited the strongest correlation

($r=0.63$), which was therefore introduced first into the ML models. Conversely, the lowest correlation was found for WT ($r=0.29$), which was included last as the 11th variable in the final combination. The predictive performance of these two models with different input combination during the testing phase is summarized in Table 5. Input combination 6 (IC-6) was found to be optimal for both models. The MLR and MLP models accounted for 65.80% and 78.70% of the variance between measured and estimated values, respectively, utilizing only six WQPs: TDS, Cl, EC, WH, TA, WS. These parameters contributed 95.77% and 91.83% of the total prediction accuracy for the respective models. Additionally, IC-6 resulted in only a small increase in RMSE values of 3.58% and 20.39%, and MAE values of 3.94% and 18.38% for the MLR and MLP models, respectively, compared to IC-11. Moreover, the artificial neural network-based MLP model showed comparatively better predictive performance than the MLR model, with an R^2 improvement of 17.88% compared to MLR. Overall, MLR and MLP models with IC-6 exhibited only a small decrease in R^2 values, 4.41% and 8.89%, respectively, compared to IC-11. This indicates that the ML models can effectively predict arsenic concentration using an optimal set of fewer, yet highly informative WQPs.

Discussion

Influence of WQPs on arsenic dynamics

Understanding the variations in WQPs is crucial for assessing water quality and potential contaminant mobilization, particularly for toxic elements like arsenic (Ibrahim et al., 2022). The mean arsenic concentration across the study area was 0.31 mg L^{-1} , substantially exceeding both the WHO guideline value of 0.01 mg L^{-1} and the Bangladesh standard of 0.05 mg L^{-1} for drinking water (ECR, 1997; Kumar & Pati, 2022). This indicates considerable public health risks. Arsenic contamination is closely linked to a hydrogeological setting where Holocene sediments rich in arsenic-bearing minerals interact with both natural weathering and human activities (Tan et al., 2020). The predominance of these sediments in the coastal region of Bangladesh facilitates the release of arsenic through reductive dissolution of iron oxyhydroxides,

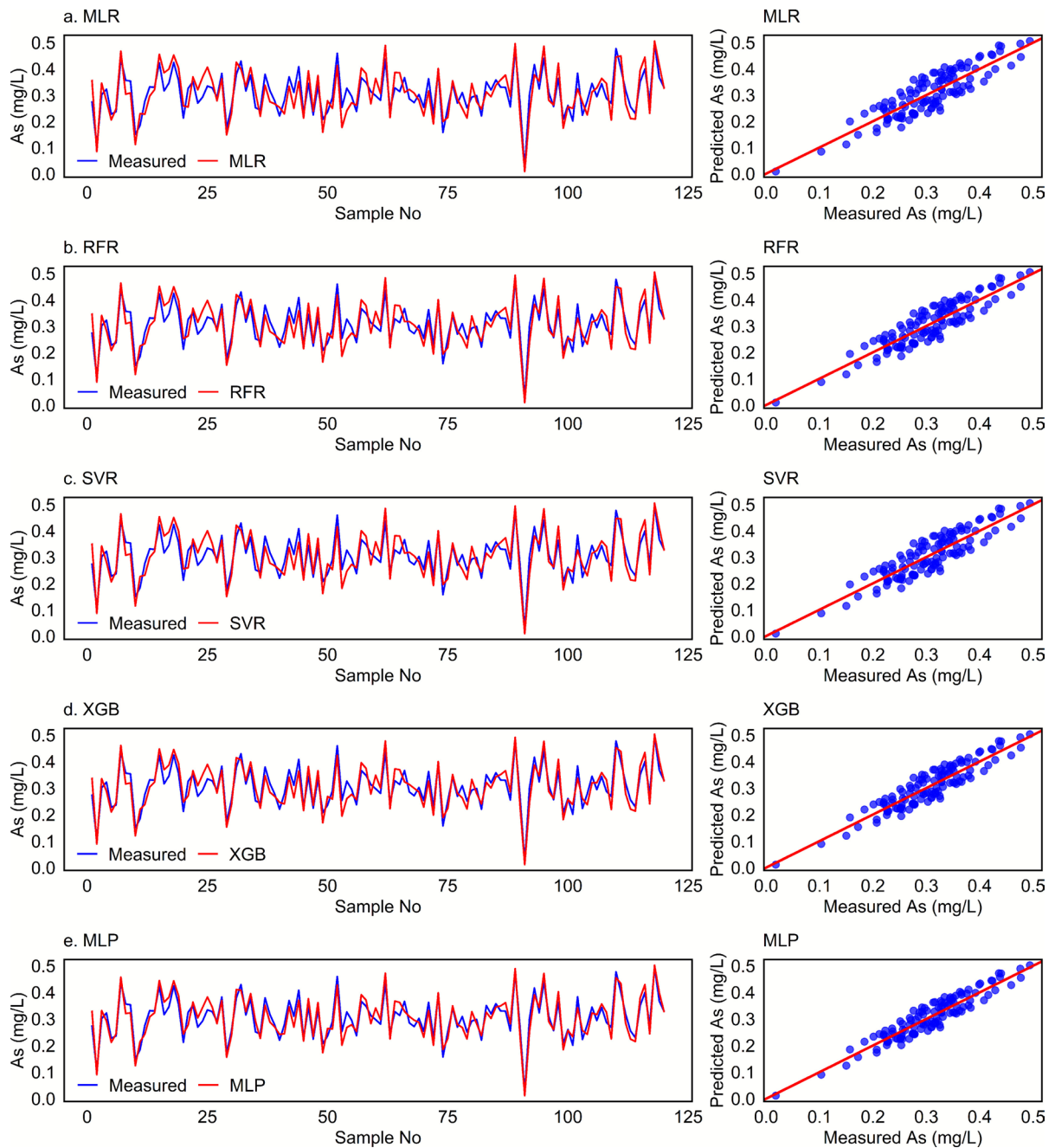


Fig. 5 Comparison of model performance between measured and predicted arsenic during testing phase across MLR, RFR, SVR, XGB, and MLP algorithms

which often serve as arsenic sinks under oxidizing conditions (Fuoco et al., 2022; Nickson et al., 2000). Additionally, the study identified significant variations in WQPs, many of which correlated with arsenic concentrations. Such variations in WQPs may

create hydrogeochemical conditions that favor the dissolution, mobilization, and persistence of arsenic in groundwater (Abiye & Bhattacharya, 2019). For example, areas experiencing seasonal saline intrusion may introduce chloride-rich water that modifies redox

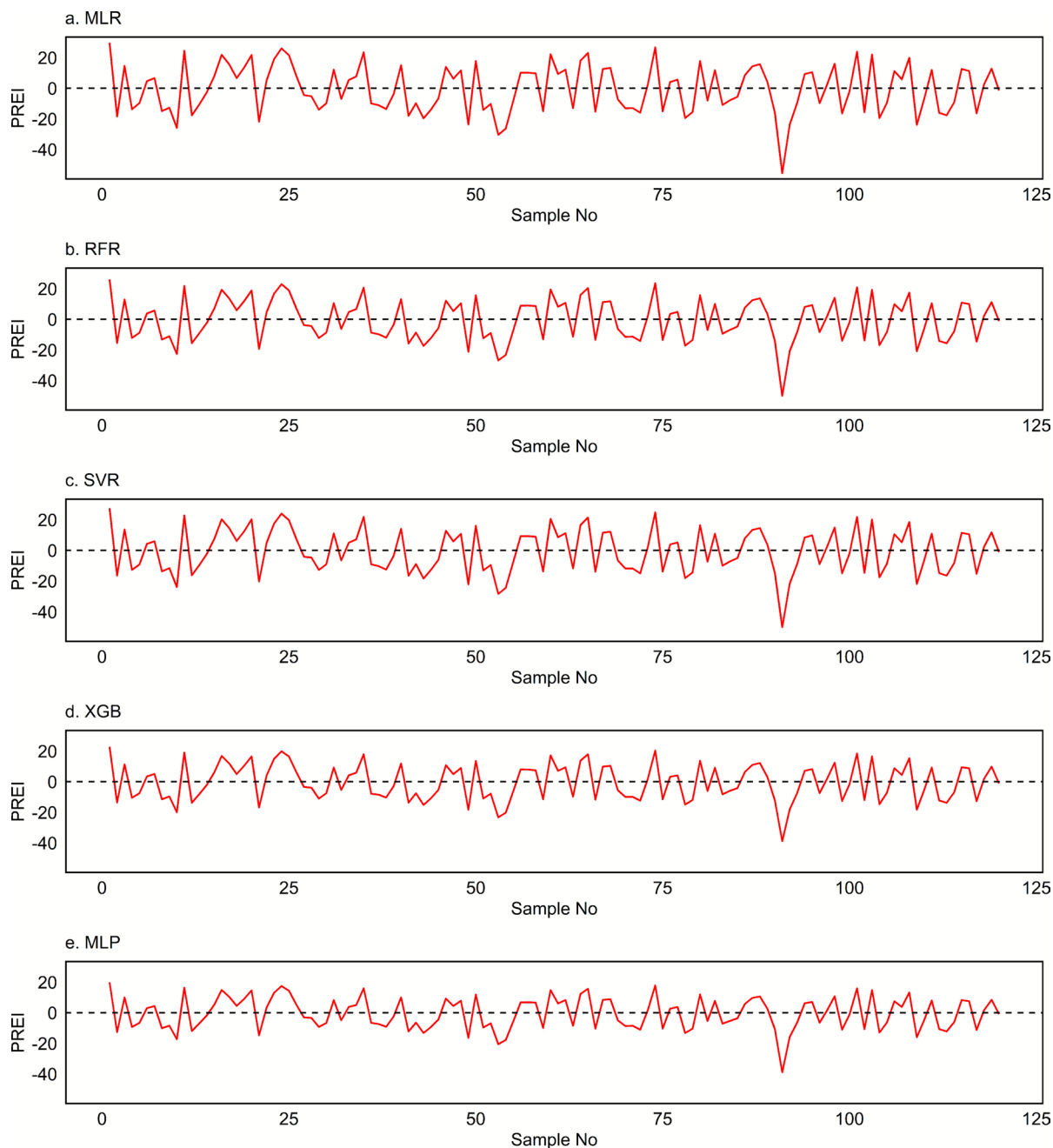


Fig. 6 Visualizations showing percentage of relative error between actual versus predicted arsenic from MLR, RFR, SVR, XGB, and MLP algorithms during testing phase

potentials and enhances the desorption of arsenic from mineral surfaces (Ayers et al., 2016). These findings indicate that arsenic contamination tends to co-occur with elevated concentrations of key dissolved constituents, and the complex interrelationships

among the WQPs indicate their combined effect on overall groundwater quality.

Among the measured parameters, WT varied between 19.14 and 26.22 °C and exhibited a weak positive correlation with arsenic ($r=0.29$), suggesting

Fig. 7 Performance evaluation of MLR, RFR, SVR, XGB, and MLP algorithms via cumulative distribution function

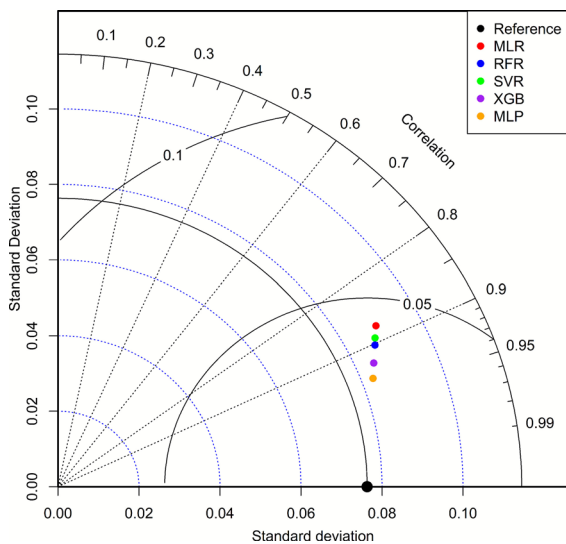
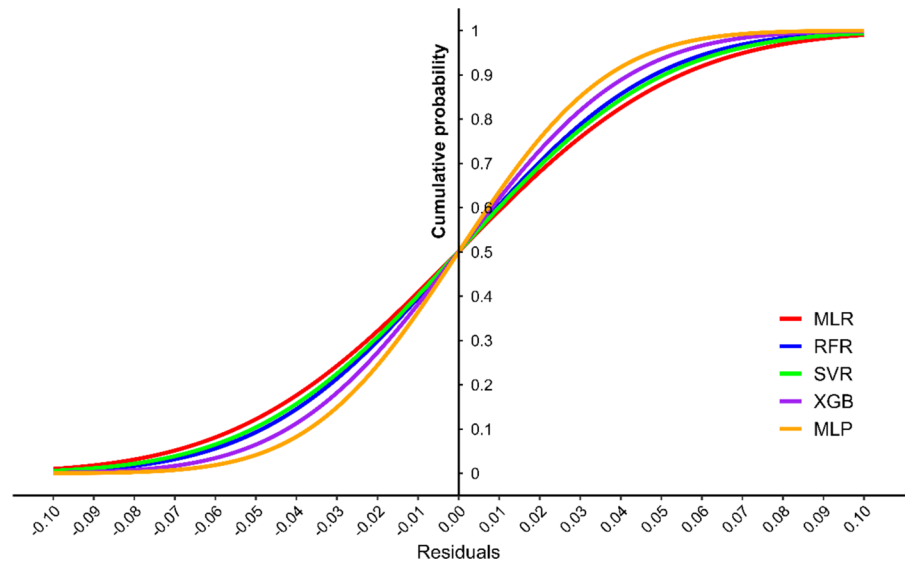


Fig. 8 Taylor diagram comparing performance of MLR, RFR, SVR, XGB, and MLP algorithms during the testing stage in predicting arsenic concentration

a potential role in mineral dissolution and weathering, which can enhance arsenic release under certain conditions (Johnston et al., 2020). WT also showed a positive correlation with TA ($r=0.33$), indicating that higher temperatures may accelerate carbonate dissolution, thereby increasing the mobility of arsenic (Bonte et al., 2013; Radu et al., 2005). Temperature-dependent enhancement of mineral weathering not only releases cations like Ca^{2+} and Mg^{2+} but also

affects the solubility of arsenate species, linking thermal regimes to arsenic bioavailability (Smedley & Kinniburgh, 2002). These interactions underscore the coupled influence of thermal and carbonate geochemistry on arsenic dynamics in groundwater systems.

EC ranged from 389 to 1355.94 $\mu\text{S cm}^{-1}$ and TDS from 195 to 693.92 mg L^{-1} , both within the permissible limits of the Bangladesh standard for drinking water ($\text{EC} \leq 1500 \mu\text{S cm}^{-1}$; $\text{TDS} \leq 1000 \text{mg L}^{-1}$) (ECR, 1997). However, significant positive correlations were observed between arsenic and EC ($r=0.55$), as well as between arsenic and TDS ($r=0.63$), demonstrating that higher dissolved mineral content supports arsenic mobility. The strong correlation between EC and TDS ($r=0.88$) indicates that ionic strength is driven by dissolved solids, which may originate from the weathering of arsenic-bearing minerals (Ahmed et al., 2010; Anawar et al., 2003), particularly from silicate and carbonate weathering (Uddin et al., 2024). The strong correlation between EC and pH ($r=0.69$) also suggests that dissolved ions and pH modulating species likely share common geological sources, such as carbonate and silicate weathering under aquifer conditions (Li et al., 2022). Similar patterns have been observed in other arsenic-affected deltas, where high ionic strength and slightly alkaline pH enhance arsenic desorption from iron oxyhydroxides (Fuoco et al., 2022). These findings indicate that, while EC and TDS values are within drinking water standards, their relationship with

Table 5 Performance of MLR and MLP models with various input combinations during the testing phase

No	Various input combinations (ICs)	MLR			MLP		
		R ²	RMSE	MAE	R ²	RMSE	MAE
1	TDS	0.402	0.059	0.054	0.431	0.058	0.052
2	TDS, Cl	0.459	0.056	0.051	0.502	0.054	0.049
3	TDS, Cl, EC	0.488	0.055	0.050	0.531	0.052	0.048
4	TDS, Cl, EC, WH	0.524	0.053	0.048	0.587	0.049	0.045
5	TDS, Cl, EC, WH, TA	0.545	0.052	0.047	0.630	0.046	0.042
6	TDS, Cl, EC, WH, TA, WS	0.658	0.045	0.041	0.787	0.035	0.032
7	TDS, Cl, EC, WH, TA, WS, DO	0.665	0.044	0.040	0.793	0.035	0.034
8	TDS, Cl, EC, WH, TA, WS, DO, COD	0.673	0.044	0.040	0.818	0.033	0.030
9	TDS, Cl, EC, WH, TA, WS, DO, COD, BOD	0.68	0.043	0.039	0.836	0.040	0.028
10	TDS, Cl, EC, WH, TA, WS, DO, COD, BOD, pH	0.683	0.043	0.039	0.848	0.030	0.027
11	TDS, Cl, EC, WH, TA, WS, DO, COD, BOD, pH, WT	0.687	0.043	0.039	0.857	0.028	0.026

arsenic implies that groundwater with elevated mineral content can serve as a medium for arsenic mobilization, even in the absence of overtly high salinity.

WH ranged between 68.88 and 323.09 mg L⁻¹, remaining within the Bangladesh standard for drinking water (≤ 500 mg L⁻¹), yet its correlation with arsenic ($r=0.53$) points to mineral dissolution processes influencing arsenic release (Rahman et al., 2023). WH also correlated with COD ($r=0.53$) and BOD ($r=0.20$), reflecting possible interactions between mineral content and organic matter. Organic matter can reduce iron and manganese oxides through microbial activity, releasing adsorbed arsenic into groundwater (Liu et al., 2023). Moreover, dissolution of calcite and dolomite releases calcium, magnesium, and potentially arsenic into groundwater (Mostafa et al., 2017). The strong correlation between WH and pH ($r=0.59$) indicates that hardness cations buffer pH, which in turn influences arsenic speciation (Jat Baloch et al., 2022). Mean TA level (568.81 mg L⁻¹) exceeded the Bangladesh recommended limit of 500 mg L⁻¹ in some locations, and its positive correlation with arsenic ($r=0.49$) suggests that alkaline conditions promote the mobilization of arsenate ions, which are more soluble at higher pH (Hafeznezami et al., 2016). High alkalinity enhances competitive desorption of arsenic from mineral surfaces and stabilizes arsenate anions, particularly in areas with high bicarbonate content (M. R. Uddin et al., 2024). Additionally, the mean pH value was 6.75, which is within the WHO recommended range of 6.5–8.5,

and showed a moderate positive correlation with arsenic ($r=0.32$). However, these slightly acidic pH conditions may have influenced arsenic mobility by promoting the dissolution of arsenic-bearing minerals and altering the surface chemistry of iron oxides, thereby facilitating arsenic release into groundwater (Katsoyiannis & Katsoyiannis, 2006; Zeng et al., 2023).

Redox conditions also played a crucial role in controlling arsenic levels (Zheng et al., 2004). DO ranged from 3.21 to 6.26 mg L⁻¹, mostly within the WHO-recommended minimum for drinking water (> 5 mg L⁻¹), but some lower values suggest localized reducing environments. A negative correlation between DO and arsenic ($r=-0.41$) indicates that oxygen depletion favors arsenic release from iron oxides and other mineral phases (Masuki et al., 2025). The weak negative correlation between DO and Cl ($r=-0.15$) suggests saline waters may be more reducing. Reducing condition conducive to arsenic mobilization via reductive dissolution of Fe and Mn oxides (Anawar et al., 2003). This mechanism has been widely documented in deltaic aquifers, where low oxygen and high salinity drive reductive dissolution, releasing both Fe²⁺ and associated arsenic (Postma et al., 2007). BOD (0.97–4.83 mg L⁻¹) and COD (49.54–184 mg L⁻¹) exceeded WHO guidelines in several samples and showed positive correlations with arsenic ($r=0.38$ and $r=0.40$, respectively), indicating that organic contamination can exacerbate arsenic release. The correlation COD and Cl ($r=0.52$) further implies that organic pollution may be associated with saline

intrusion, likely from agricultural runoff or wastewater (Liu et al., 2022).

Salinity ranged from 0.17 to 2.31 PSU, remaining below the Bangladesh drinking water threshold (≤ 5 PSU), however, its positive correlation with arsenic ($r=0.45$) indicates saline water intrusion as a contributing factor to arsenic mobilization through enhanced mineral dissolution (Parvaiz et al., 2020; Zhi et al., 2022). The strong correlation between WH and salinity ($r=0.76$) indicates the common coexistence of salinity and hardness. Areas with higher salinity often experience elevated arsenic due to increased dissolution of arsenic bearing minerals (Jia et al., 2017). In other coastal regions, the combination of salinity intrusion and shallow aquifer conditions has been shown to increase arsenic release through both ion exchange and reductive dissolution pathways (Nickson et al., 2000; M. R. Uddin et al., 2024). Collectively, these relationships indicate that even when most WQPs are within national or WHO guideline values, their interactions, particularly under conditions of higher mineral content, alkalinity, salinity, and organic load, can create hydrogeochemical environments conducive to arsenic mobilization.

Comparative performance of various ML models for arsenic prediction

Accurate prediction of arsenic concentrations is crucial for environmental monitoring and health risk assessment due to its severe hazards (Ibrahim et al., 2022). Thus, a comparative evaluation of machine learning models, such as MLR, RFR, SVR, XGB, and MLP, was conducted to assess their suitability in predicting arsenic concentrations in groundwater. Among the evaluated models, MLP consistently demonstrated superior performance in both training and testing phases. Scatter plots (Fig. 5) showed that MLP predictions closely aligned with the 1:1 reference line, while the PREI analysis (Fig. 6) indicated minimal prediction uncertainty. Additionally, the cumulative distribution analysis (Fig. 7) revealed that MLP had the highest proportion of residuals within a narrow error range, and the Taylor diagram (Fig. 8) demonstrated lower RMSE with higher correlation compared to the other models. MLP's ability to model complex, nonlinear relationships through hidden-layer feature extraction makes it particularly well-suited for high-dimensional water quality datasets,

unlike MLR, which assumes linearity (Najah et al., 2013). While XGB and RFR also handle nonlinearities, their reliance on decision trees limits their capacity to model intricate interactions (S. Uddin & Lu, 2024). SVR, though effective in specific regression tasks, requires intensive tuning and may struggle to generalize across diverse datasets (Zhou et al., 2024). MLP's flexible architecture, optimized through backpropagation, minimizes overfitting and adapts to diverse environmental datasets, making it the most reliable model for groundwater arsenic prediction (Wan et al., 2018). The overall performance ranking was $MLP > XGB > RFR > SVR > MLR$ for predicting arsenic concentrations in groundwater.

Previous studies have demonstrated the effectiveness of ML approaches for arsenic prediction, but their performance has varied widely across regions. In the present study, the MLP explained 86% of the variance in arsenic concentrations using only hydrochemical WQPs (Table 4), outperforming several earlier statistical and hybrid models that relied on broader, multi-domain predictors. For example, ANFIS- and regression-based global models explained only 65% and 56–59% of arsenic variability, respectively (Amini et al., 2008), while logistic regression-based models in Bangladesh achieved just 72.18% accuracy (Zhao et al., 2024). Even the boosted regression tree approach using three-dimensional groundwater parameters explained 71–82% variance in the U.S. Glacial Aquifer System (Erickson et al., 2021), which is still lower than the performance of the current MLP despite its more concise input structure. Compared with high-performing ML regressors, (Ibrahim et al., 2022) reported > 93% variance explained by XGB, LGB, and GRNN in Ghana, though those models used extensive input variables without evaluating optimal combinations. In contrast, the present MLP achieved 86% variance explained with hydrochemically relevant predictor set alone, reflecting an efficient balance between complexity and predictive accuracy. Similarly, RFR models applied in the Hetao Basin explained up to 80% variance (Zhao et al., 2024), but in this study area, the MLP outperformed RFR and XGB, indicating that coastal hydrochemical interactions favor nonlinear neural architectures. The weaker performance of linear models in this study ($R^2=0.69$) also aligns with previous findings that MLR tends to underestimate arsenic in redox-dominated aquifers. Other regional studies also emphasize the limitations

of linear approaches. For example, Akajiaku et al., (2025) reported that while MLR achieved high accuracy for predicting iron concentrations ($R^2=0.92$), its generalizability was constrained by the small sample size ($n=50$) and the absence of key hydrogeological variables. Although hybrid PC-ANN and RFR-UFS frameworks have shown improved predictive ability (Bindal & Singh, 2019; Cao et al., 2018), their performance remains below the current MLP when evaluated strictly in terms of variance explained. The overall comparisons show that ML model performance across arsenic studies ranges widely from 59 to >93%, depending on predictor types and model architecture. The strong performance of the present MLP highlights the importance of using mechanistically relevant hydrochemical variables rather than proxy geological and spatial predictors. However, the performance of ML algorithms may still vary considerably across regions due to differences in dataset characteristics and the choice of input parameters (Ibrahim et al., 2022). Thus, there is no universally “best” or “most robust” algorithm across all scenarios (Bui et al., 2020). Model selection must be intended for specific contexts and regions through comprehensive evaluations to identify the most reliable model for predicting arsenic concentrations.

Additionally, selecting optimal input combinations (ICs) is crucial for enhancing ML model performance (Terzi et al., 2023), as it focuses on the most influential variables, optimizing predictive power while avoiding complexity (Lygeros & Sutter, 2016). The effect of ICs varies across regions due to spatial heterogeneity and nonlinear interactions among WQPs caused by diverse pollution sources (Bui et al., 2020). Including too many variables in an ML model can sometimes reduce predictive accuracy, as redundant or irrelevant predictors introduce noise and multicollinearity, potentially obscuring the true relationships between input variables and arsenic concentrations (Terzi et al., 2023). This study identified IC-6, comprising TDS, Cl, EC, WH, TA, and WS, as optimal for both MLR and MLP models. These six parameters are hydrochemically significant because they collectively capture the key geochemical conditions controlling arsenic mobilization: TDS and EC reflect the overall ionic strength and mineralization of groundwater, influencing arsenic solubility (Anawar et al., 2003); Cl indicates chloride levels, which can compete with arsenic for adsorption sites (Nasher, 2022);

WH and TA represent water hardness and alkalinity, which affect carbonate equilibria and buffering capacity, modulating arsenic release (Rahman et al., 2023; Uddin et al., 2024); and WS captures overall salinity, which can drive desorption of arsenic from iron hydroxides (Fuoco et al., 2022; Nickson et al., 2000; Uddin et al., 2024). Together, they provide a concise yet comprehensive representation of the processes that govern arsenic behavior in the southeastern coastal aquifers of Bangladesh. Using IC-6, MLR and MLP explained 65.80% and 78.70% of the variance, contributing 95.77% and 91.83% to total prediction accuracy, respectively (Table 5). Moreover, the MLP outperformed MLR by 17.88% in R^2 , reflecting its superior ability to model complex nonlinear relationships compared to MLR (Basak et al., 2025). Overall, both models effectively predicted arsenic concentrations using a reduced yet informative set of WQPs. However, for the southeastern coastal region of Bangladesh, which is highly vulnerable to arsenic contamination due to its unique hydrogeological setting and increasing anthropogenic pressures on groundwater resources, MLP’s capacity to capture complex nonlinear interactions may make it particularly suitable for accurately predicting arsenic concentration.

Implications of this study and policy recommendations

The findings of this study provide crucial insights for effective groundwater quality management in data-scarce and resource-limited regions like the southeastern coast of Bangladesh. By employing five ML models, this research demonstrated the feasibility of arsenic prediction using an optimal set of key WQPs, offering a cost-effective monitoring solution that can be readily incorporated into existing groundwater-quality programs. The strong performance of the MLP model, despite the reduced number of input variables, indicates that local authorities and communities could deploy this model using routinely measurable parameters, enabling rapid screening and near real-time assessment of arsenic risks. Instead of measuring numerous physicochemical, biological, climatic, geological, and spatial parameters, monitoring efforts can focus on the optimized IC-6 set (TDS, Cl, EC, WH, TA, and WS), which reliably predicts arsenic while reducing laboratory and equipment costs. This

targeted approach offers clear advantages over conventional methods by lowering operational burdens, improving efficiency, and allowing early detection of contamination, particularly in regions where financial and technical constraints limit comprehensive testing. Academic institutions, local authorities, and communities should therefore collaborate to integrate ML-based monitoring frameworks and optimized ICs into practical groundwater-quality assessment and management strategies.

Conclusions

This study demonstrated the strong potential of ML algorithms for predicting groundwater arsenic concentrations in the southeastern coastal region of Bangladesh using a set of key WQPs. Arsenic occurrence showed strong associations with salinity-driven parameters, particularly TDS, Cl, and EC, and the mean concentration (0.31 mg L^{-1}) exceeded both WHO and national standards, indicating a potential health risk. Among the evaluated models, the MLP consistently outperformed MLR, RFR, SVR, and XGB, and the optimal input combination (IC-6: TDS, Cl, EC, WH, TA, WS) allowed the MLP to achieve high accuracy. With IC-6, the MLP model explained 78.7% of the variance and 91.83% of the total prediction accuracy in predicting arsenic concentration. This indicates that reliable arsenic prediction can be achieved using fewer yet highly informative WQPs, reducing model complexity. These findings extend earlier research by showing that reliable arsenic prediction is possible with fewer but hydrochemically meaningful variables. The study also highlights practical implications for resource-limited regions where full-scale water testing is difficult. However, the analysis did not include factors such as well depth, seasonal shifts, hydrogeological conditions, or spatial variability, which may influence model performance. These limitations should be considered in future studies to improve predictive outcomes. Overall, the study offers a cost-effective tool for early identification of arsenic-prone wells. Researchers and policy-makers are encouraged to incorporate hydrogeological data, expand spatial coverage, and test hybrid ML approaches to enhance model generalizability across diverse arsenic-affected regions.

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Data availability The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no financial or personal conflicts of interest that could have influenced this work.

Ethical responsibilities of authors All authors have read, understood, and have complied as applicable with the statement on "Ethical responsibilities of Authors" as found in the Instructions for Authors.

Ethical approval for animal research This study did not involve any experiments on animals.

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